STAT 940

Deep Learning

Term: Fall 2026
Class meetings: Tuesdays and Thursdays, 1:00-2:20 p.m.
Location: MC 4063
Instructor: Ali Ghodsi
Instructor office: M3 4009
Instructor office hours: Wednesdays, 1:00-2:00 p.m. - course concepts and final project
Teaching assistant: To be announced
TA office hours: To be announced - assignments, code, and technical questions

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Course description

This graduate course develops mathematical, algorithmic, and computational understanding of modern deep learning. It emphasizes theoretical foundations, model analysis, practical implementation, and critical engagement with current research.

Prerequisites

Students are expected to have prior coursework or equivalent background in machine learning, probability, calculus, linear algebra, and statistics. Programming experience is also expected for computational assignments and the final project.

Course format

This course is delivered in person.

Materials and resources

Required textbook

Benyamin Ghojogh and Ali Ghodsi, **Elements of Deep Learning**, Springer, 2026. ISBN 978-3-032-10738-1 (eBook).

Book page: https://link.springer.com/book/10.1007/978-3-032-10738-1

Assigned readings will be specified by chapter and section. The lectures define the course curriculum; the course does not cover the textbook sequentially or in its entirety.

Additional reference

Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.

Online edition: https://www.deeplearningbook.org/

Course platforms

Platform	Primary role
Course website	Course materials, readings, schedules, instructions, and resources.
Piazza	Course communication, questions, and private feedback.
Learn	Dropboxes and grades.

Course website: Go to https://aghodsib.github.io/, select Teaching, and then select Deep Learning.

Piazza: Access Piazza through the widget on the course website.

Students are responsible for checking all official course platforms regularly.

Communication policy

All course communication with the instructor and teaching assistant must use Piazza: public posts for class-wide questions and private posts for personal matters or private feedback. Do not email the instructional team directly about course matters.

Assessment and evaluation

Component	Description	Weight
Assignments	Three individual assignments at 15% each	45%
Paper presentation	Up to four students	10%
Group final project	Groups of up to four students	45%
Total		**100%**

Assignments - 45%

There will normally be three individual assignments, each worth 15% and released two weeks before its due date. Assignments may include derivations, proofs, programming, and experiments.

To confirm authorship and understanding, a student may be required to explain part of a submission orally or complete a short individual quiz based on it. Inadequate demonstrated understanding may result in reduced or no credit for the affected work.

Paper presentation - 10%

A paper presentation may have up to four students. The presentation group and final-project group may be the same or different. Papers must be selected from the eligible venues: NeurIPS, ICML, ICLR, AISTATS, AAAI, IJCAI, UAI, CVPR, ICCV, ECCV, ACL, EMNLP, NAACL, JMLR, TMLR, IEEE TPAMI, or Artificial Intelligence. Each member must participate and answer questions.

Group final project - 45%

Groups of up to four students submit one final report. Detailed requirements will be provided. The project must be new work for STAT 940 and may not reuse pre-existing thesis work or work submitted for another course.

Tentative assessment dates

Assessment	Release	Due
Assignment 1	Monday, October 5	Monday, October 19, 11:59 p.m.
Assignment 2	Monday, November 2	Monday, November 16, 11:59 p.m.
Assignment 3	Monday, November 23	Monday, December 7, 11:59 p.m.
Paper presentation	Eligible venues listed above	Scheduled presentation date
Group final project	To be announced	Monday, December 21, 11:59 p.m.

All dates are tentative and may change. Changes will be announced.

Late and missed work

Written assignments and final project

Work submitted within 24 hours after the deadline receives a 10% penalty on the earned mark; after 24 hours, the grade is zero.

Generative artificial intelligence

Generative AI may support learning unless an assessment states otherwise, but submitted work must reflect the student's own authorship and understanding. Substantive use must be disclosed, and students must verify all work, citations, code, and results. Generative AI is prohibited during oral or quiz-based verification. Unauthorized or undisclosed use may constitute an offence under Policy 71.

Tentative course plan and textbook alignment

Tentative topic	Textbook alignment
Foundations, perceptron, and feedforward neural networks	Ch. 1, Secs. 2-3 and 5; Ch. 2, Secs. 2.2-2.9 and 4
Backpropagation and optimization	Ch. 3, Secs. 2-5
Model selection, regularization, and generalization	Ch. 5, Secs. 4-11; Ch. 6, Secs. 2-3
Convolutional neural networks	Ch. 4, Secs. 2-3 and 5.2-5.7
Recurrent neural networks and sequence-to-sequence models	Ch. 8, Secs. 2-5; Ch. 9, Secs. 2.1-2.3
Attention, Transformers, and Performer	Ch. 9, Secs. 2.3-4.2
Large language models and alignment	Ch. 11, Secs. 2-4 and 7-9
Variational models, moment matching, and GANs	Ch. 12, Secs. 2-3; Ch. 13, Secs. 2-5; Ch. 14, Secs. 2-3
Diffusion models	Ch. 15, Secs. 2-4
Graph neural networks	Ch. 17, Secs. 2-7
Deep reinforcement learning, if time permits	Ch. 18, Secs. 2-9
Presentations and selected emerging topics	Assigned papers; no fixed textbook section

Assignment screening and Turnitin

Turnitin may be used to check source documentation. Submissions are stored on a U.S. server. Students will receive notice of its use and available alternatives; requests for an alternative must be made in the first week or when the relevant assignment details are provided.

Waterloo Turnitin information: https://uwaterloo.ca/academic-integrity/integrity-students/turnitin-and-ithenticate

Academic integrity and University policies

Academic integrity

Waterloo community members are expected to promote honesty, trust, fairness, respect, and responsibility.

Office of Academic Integrity: https://uwaterloo.ca/academic-integrity/

Ideas, text, code, data, figures, and other intellectual property must be properly attributed. Plagiarism and other misconduct will be addressed under University policy.

Grievance

Students who believe a University decision was unfair or unreasonable should consult Policy 70 or the department's administrative assistant.

Policy 70: https://uwaterloo.ca/secretariat/policy-70-student-petitions-grievances-and-requests

Discipline

Students are responsible for understanding academic-integrity requirements. Questions about plagiarism, cheating, collaboration, or group work should be directed to the instructor, academic advisor, or appropriate associate dean.

Policy 71, Student Discipline: https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71

Appeals

Appeals of eligible decisions or penalties under Policies 70 and 71 are governed by Policy 72.

Policy 72, Student Appeals: https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-72

Accessibility Services

Waterloo provides academic accommodations consistent with the Ontario Human Rights Code. Students who may require accommodation should register with AccessAbility Services promptly; registered students must activate accommodations for this course.

AccessAbility Services: https://uwaterloo.ca/accessability-services/

Contact: 519-888-4567, ext. 35082; access@uwaterloo.ca; Needles Hall North, first floor, Room 1401.

Mental health supports

Confidential counselling, workshops, and crisis support are available through Counselling Services.

Counselling Services: https://uwaterloo.ca/campus-wellness/counselling-services/

Phone: 519-888-4096.

Technical support

For Learn problems, contact learnhelp@uwaterloo.ca with your name, WatIAM ID, student number, and course. Support hours are Monday-Friday, 8:30 a.m.-4:30 p.m. Eastern Time.

Course recording notice

This course may be audio and video recorded. Recordings may capture students' images, voices, questions, or other classroom participation and may be distributed through University platforms and publicly accessible platforms, including YouTube.

Students who do not wish their identifiable image or voice to appear in published recordings must notify the instructor by **September 15, 2026**, at **ali.ghodsi@uwaterloo.ca**, using the subject line **[STAT 940 Recording Privacy]**, so that appropriate arrangements can be made.